

Task 1:

```
#include <iostream>
```

```
#include <fstream>
```

```
#include <string>
```

```
using namespace std;
```

```
bool isLetter(char c){  
    return ((c>='A' && c<='Z') || (c>='a' && c<='z'));  
}
```

```
bool isDigit(char c){  
    return ((c>='0' && c<='9'));  
}
```

```
bool isIdentifier(string s){  
    if(s.length()==0){  
        return false;  
    }
```

```
    if(!(isLetter(s[0]) || s[0]=='_')){  
        return false;  
    }
```

```
    for(int i=1;i<(int)s.length();i++){  
        if(!(isLetter(s[i]) || isDigit(s[i]) || s[i]=='_')){  
            return false;  
        }  
    }  
}
```

```
    return true;
}
```

```
bool isVariable(string type,string id,char last){
    if((type=="int" || type=="float" || type=="double" || type=="char") && last==';' && isIdentifier(id)){
        return true;
    }
    return false;
}
```

```
bool isFunction(string type,string fname){
    if((type=="int" || type=="float" || type=="double" || type=="char" || type=="void") &&
isIdentifier(fname)){
        return true;
    }
    return false;
}
```

```
string removeSpaces(string s){
    string temp="";

    for(int i=0;i<(int)s.length();i++){
        if(s[i]!=' '){
            temp=temp+s[i];
        }
    }

    return temp;
}
```

```
}
```

```
int main()
```

```
{
```

```
    ofstream fout("input.txt");
```

```
    int n;
```

```
    cout<<"How many lines: ";
```

```
    cin>>n;
```

```
    cin.ignore();
```

```
    string line;
```

```
    for(int i=0;i<n;i++){
```

```
        cout<<"Enter line "<<i+1<<": ";
```

```
        getline(cin,line);
```

```
        fout<<line<<endl;
```

```
    }
```

```
    fout.close();
```

```
    ifstream fin("input.txt");
```

```
    ofstream report("report.txt");
```

```
    ofstream valid("valid_code.txt");
```

```
    int total=0;
```

```
    int varCount=0;
```

```
    int funcCount=0;
```

```
int invalidCount=0;
```

```
while(getline(fin,line)){
```

```
    total++;
```

```
    string originalLine=line;
```

```
    string type,name,rest;
```

```
    int i=0;
```

```
    while(i<(int)line.length() && line[i]!=' '){
```

```
        i++;
```

```
    }
```

```
    while(i<(int)line.length() && isLetter(line[i])){
```

```
        type=type+line[i];
```

```
        i++;
```

```
    }
```

```
    while(i<(int)line.length() && line[i]!=' '){
```

```
        i++;
```

```
    }
```

```
    while(i<(int)line.length()){
```

```
        rest=rest+line[i];
```

```
        i++;
```

```
    }
```

```
    rest=removeSpaces(rest);
```

```

if(type==""){
    report<<"Line "<<total<<": Invalid"<<endl;
    invalidCount++;
    continue;
}

if(rest.length()>0 && rest.find('(')==-1 && rest[rest.length()-1]==';'){
    name=rest.substr(0,rest.length()-1);

    if(isVariable(type,name,rest[rest.length()-1])){
        report<<"Line "<<total<<": Variable Declaration"<<endl;
        valid<<originalLine<<endl;
        varCount++;
    } else{
        report<<"Line "<<total<<": Invalid"<<endl;
        invalidCount++;
    }
} else if(rest.find('(')!=-1){
    int p=rest.find('(');
    name=rest.substr(0,p);

    if(rest.substr(p)=="()"){
        if(isFunction(type,name)){
            report<<"Line "<<total<<": Function Declaration"<<endl;
            valid<<originalLine<<endl;
            funcCount++;
        } else{
            report<<"Line "<<total<<": Invalid"<<endl;

```

```

        invalidCount++;
    }
} else{
    report<<"Line "<<total<<" : Invalid"<<endl;
    invalidCount++;
}
} else{
    report<<"Line "<<total<<" : Invalid"<<endl;
    invalidCount++;
}
}

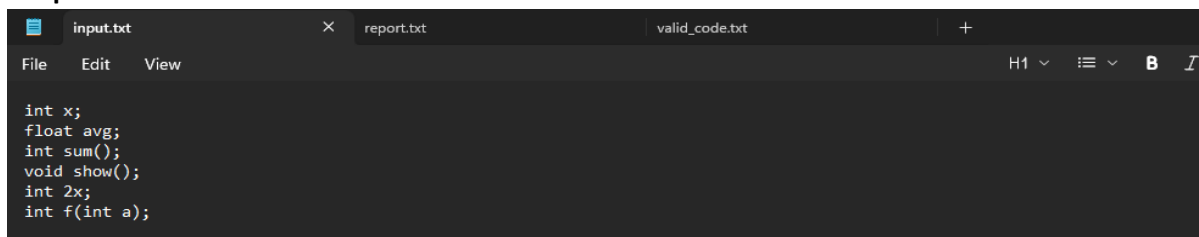
report<<"Summary:"<<endl;
report<<"Total lines = "<<total<<endl;
report<<"Variable Declarations = "<<varCount<<endl;
report<<"Function Declarations = "<<funcCount<<endl;
report<<"Invalid = "<<invalidCount<<endl;

fin.close();
report.close();
valid.close();

return 0;
}

```

Output:



```

input.txt
report.txt
valid_code.txt

File Edit View H1  B I

int x;
float avg;
int sum();
void show();
int 2x;
int f(int a);

```

```
input.txt  report.txt  valid_code.txt
File Edit View H1 H2 H3 H4 H5 H6 B I
Line 1: Variable Declaration
Line 2: Variable Declaration
Line 3: Function Declaration
Line 4: Function Declaration
Line 5: Invalid
Line 6: Invalid

Summary:
Total lines = 6
Variable Declarations = 2
Function Declarations = 2
Invalid = 2
```

```
input.txt  report.txt  valid_code.txt
File Edit View H1 H2 H3 H4 H5 H6 B I

int x;
float avg;
int sum();
void show();
```